

Cloud LearnX

Mathematics

Transformations and Grid geometry — Practice Worksheet

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Topic: Transformations and Grid geometry

Instructions:

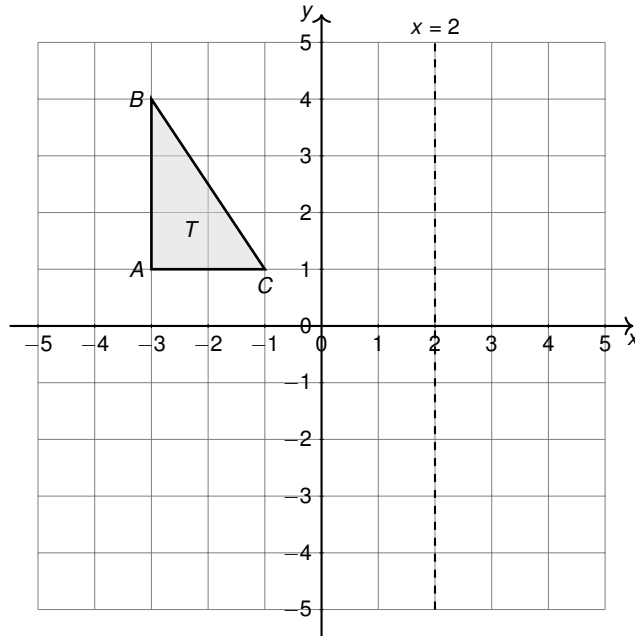
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** Triangle T is drawn on the grid below. The line $x = 2$ is also shown.

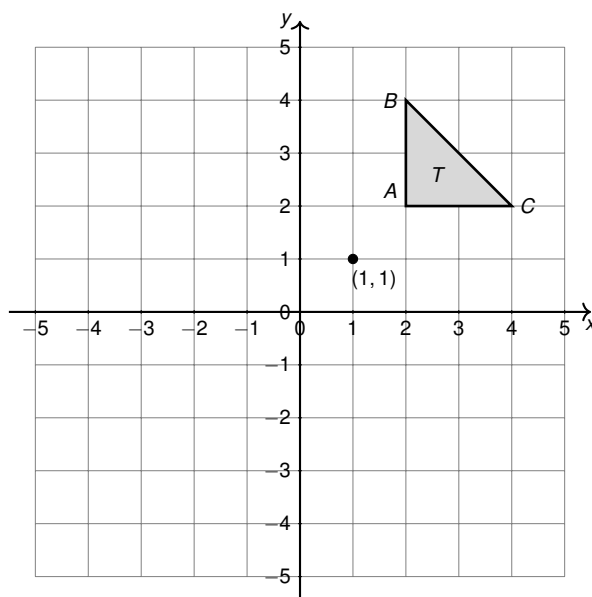


Reflect triangle T in the line $x = 2$.

(2)

(Total for Question 1 is 2 marks)

2 Shape T is drawn on the grid below. The point marked with a dot is the centre of rotation, $(1, 1)$.



Rotate shape T 90° clockwise about the point $(1, 1)$.

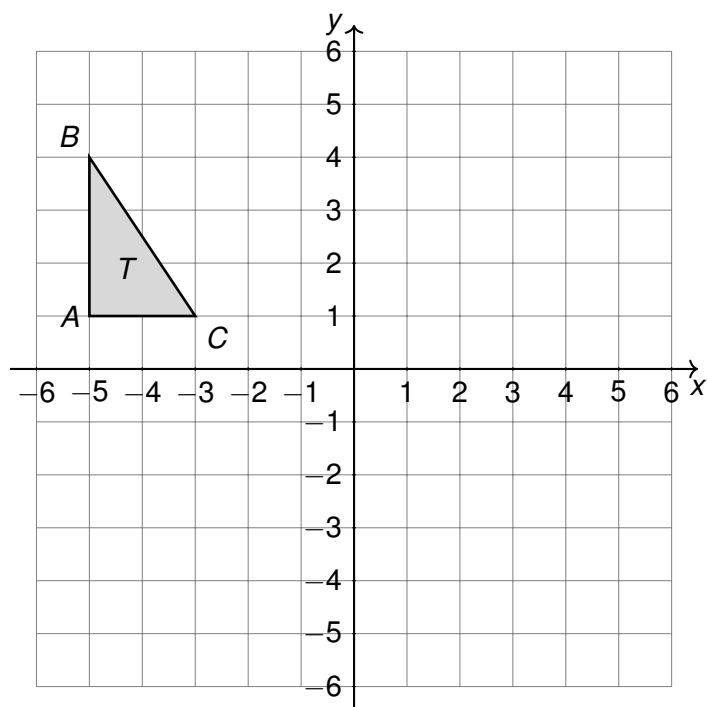
Draw the image of shape T on a grid and label it T' , stating the coordinates of the image of each vertex.

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(3)

(Total for Question 2 is 3 marks)

3 Triangle T is drawn on the grid below.



Triangle T is translated by the vector $\begin{pmatrix} 6 \\ -2 \end{pmatrix}$ to give triangle T' .

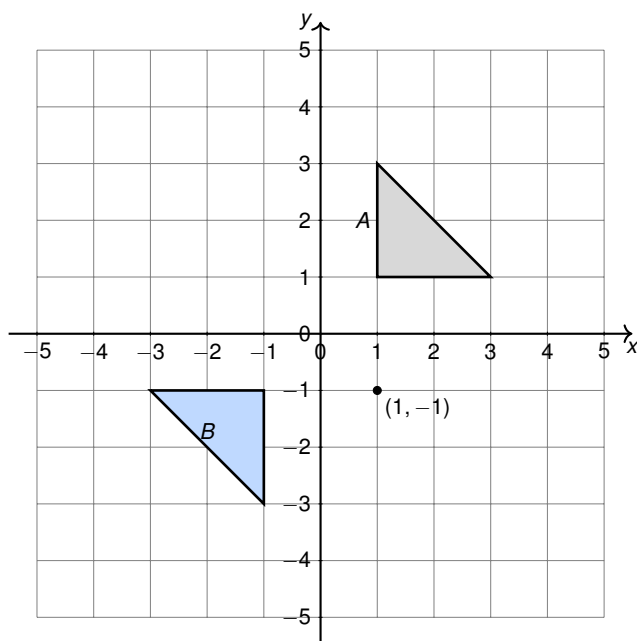
On the grid, draw and label triangle T' , and state the coordinates of the image of vertex A .

(..... ,)

(2)

(Total for Question 3 is 2 marks)

4 Shape A and shape B are drawn on the grid below.



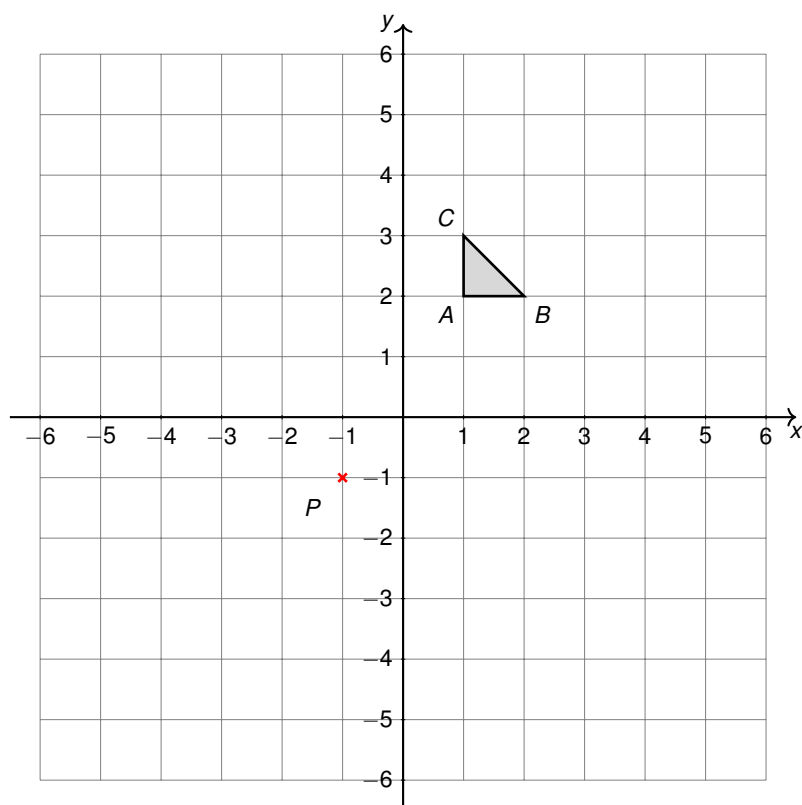
Describe fully the single transformation that maps shape A onto shape B .

(3)

(Total for Question 4 is 3 marks)

5 Triangle T has vertices at $A(1, 2)$, $B(2, 2)$ and $C(1, 3)$.

The point $P(-1, -1)$ is the centre of enlargement.



Enlarge triangle T by scale factor 2, centre $P(-1, -1)$.

(a) Work out the coordinates of the image of each vertex of triangle T .

.....

(3)

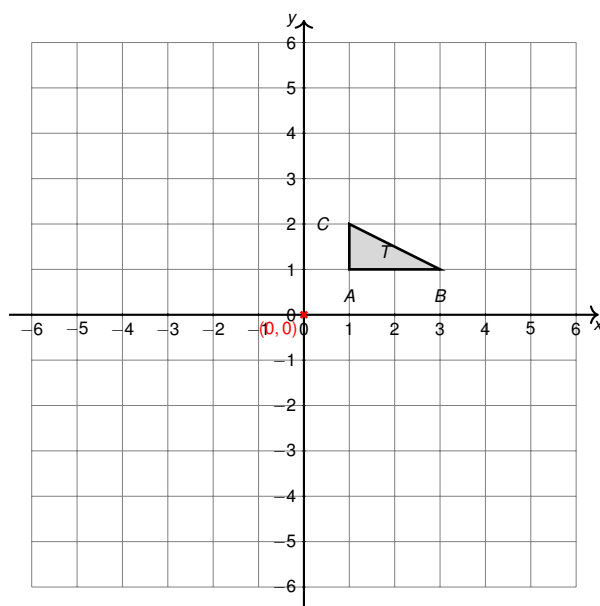
(b) On the grid above, draw the enlarged triangle.

(1)

(Total for Question 5 is 4 marks)

6 Triangle T has vertices $A(1, 1)$, $B(3, 1)$ and $C(1, 2)$.

Triangle T is enlarged by scale factor -2 , centre $(0, 0)$, to give triangle T' .



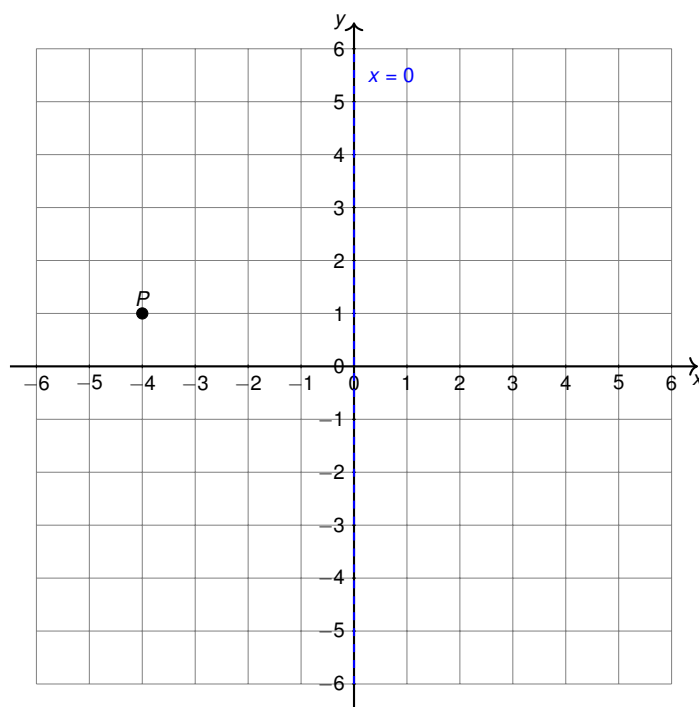
Find the coordinates of the vertices of triangle T' , the image of T after this enlargement.

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(4)

(Total for Question 6 is 4 marks)

- 7 A research satellite tracks a rare specimen on a grid map of a nature reserve. At 06:00, the specimen is marked at point $P(-4, 1)$ on the grid.



Between 06:00 and 06:10, the specimen moves according to the column vector $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$, taking it to point Q .

Between 06:10 and 06:20, the satellite's tracking system then reflects the recorded position in the line $x = 0$ (the y -axis), giving the final logged position R .

- (a) Find the coordinates of point Q , the specimen's position at 06:10.

(..... ,)

(2)

- (b) Show that the final logged position R is $(-1, -2)$.

(2)

(Total for Question 7 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS